

Part B Question 5

Algorithm:- Take input for total number of cars, Initialize all the counter and variables for capacity. Take use input for vehicle type, user category and permit and check their validity. Check if vehicle is emergency or not and check the zone capacity to allow the specific zone if space available and update the capacity output all the summary.

PAC:-

Given Data	Process	Required Results
- No. of vehicles - vehicle type - permit - Emergency	- Validate inputs - Check Emergency - Allow the zone - Check capacity - Update the capacities - Update the counters	- Assigned zone - Remaining capacity - Rejected vehicle - Accepted vehicles - Capacities

IPO chart:-

Inputs	Process	OUTPUTS
- N - Vt - Permit - Emergency	- Validate inputs - Verify Emergency - Allow the zone - Check capacity - Update all variables	- No. of vehicles process - Accepted vehicles - Rejected vehicle - No. of bikes cars and vans - Remaining capacity - Zone with highest occupancy

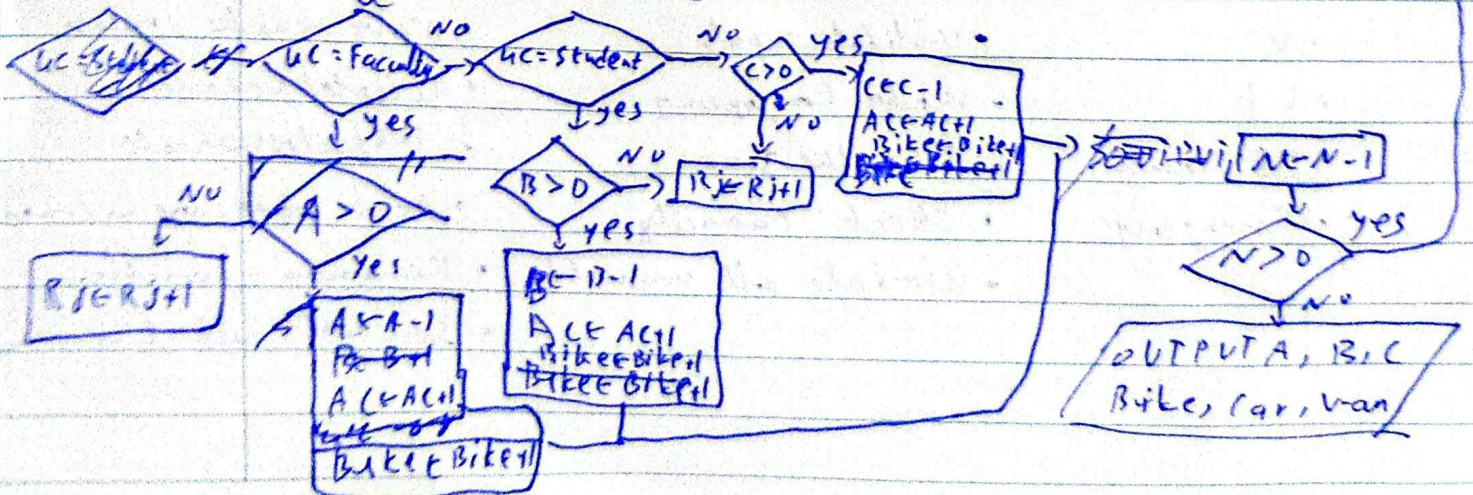
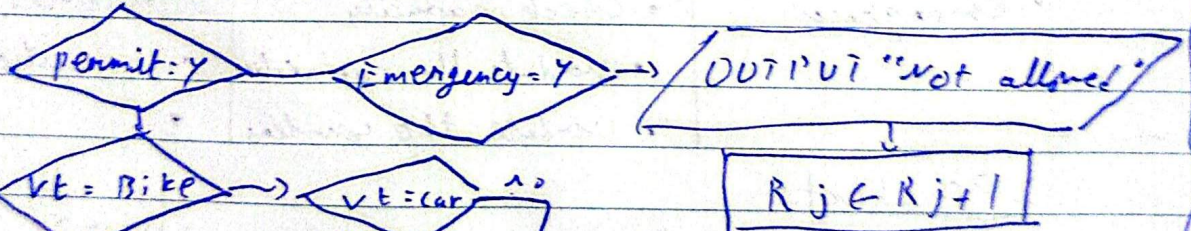
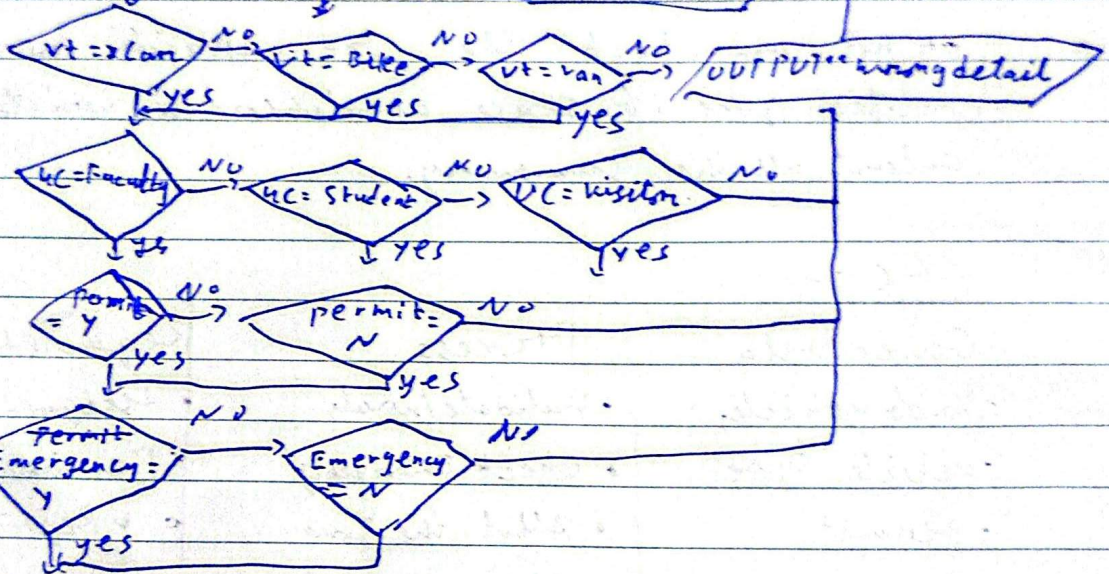
Flow Chart:

Start

A ← 20 B ← 40 C ← 15
N ← 0 Acc 0 Car 0 Bike 0 Van 0

INPUT N

INPUT Vt, UC, Permit, Emergency



Pseudocode :-

INPUT $N \rightarrow A \leftarrow 0, R_j \leftarrow 0, \text{Bike} \leftarrow 0, \text{Car} \leftarrow 0, \text{Van} \leftarrow 0, A \leftarrow 20, B \leftarrow 40, C \leftarrow 15$
 WHILE $N > 0$ DO

~~INPUT~~ $vt, UC, \text{Permit}, \text{Emergency}$

~~Details~~ $\leftarrow$ ~~TRUE~~ FALSE

WHILE $\text{Details} = \text{FALSE}$ DO

$\text{Details} \leftarrow \text{TRUE}$

IF $vt \neq \text{Bike}$ AND $vt \neq \text{Car}$ AND $vt \neq \text{Van}$ THEN

$\text{Details} \leftarrow \text{FALSE}$

ENDIF

IF $UC \neq \text{Fac}$ AND $UC \neq \text{Stud}$ AND $UC \neq \text{Visit}$ THEN

$\text{Details} \leftarrow \text{FALSE}$

ENDIF

IF $\text{Permit} \neq Y$ AND $\text{Permit} \neq N$ THEN

$\text{Details} \leftarrow \text{FALSE}$

ENDIF

IF $\text{Emergency} \neq Y$ AND $\text{Emergency} \neq N$ THEN

$\text{Details} \leftarrow \text{FALSE}$

ENDIF

ENDWHILE

IF $vt = \text{Bike}$ THEN IF $\text{Permit} = N$ THEN

IF $UC = \text{Fac}$ THEN $R_j \leftarrow R_j + 1$

IF $A > 0$ THEN ELSE IF $\text{Emergency} = N$ THEN

$A \leftarrow A - 1$

$R_j \leftarrow R_j + 1$

$\text{Bike} \leftarrow \text{Bike} + 1$ ELSE

$A \leftarrow A + 1$

ENDIF ELSE

~~ENDIF~~ $R_j \leftarrow R_j + 1$

ENDIF

ENDIF

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IF IF vt = car THEN
    IF
    IF vt = Bike THEN
        IF uc = Fac THEN
            IF A > 0 THEN
                A ← A - 1
                Bike ← Bike + 1
                Ac ← Ac + 1
            ELSE
                Rj ← Rj + 1
            ENDIF
        ELSE IF uc = stud THEN
            IF B > 0 THEN
                B ← B - 1
                Bike ← Bike + 1
                Ac ← Ac + 1
            ELSE
                Rj ← Rj + 1
            ENDIF
        ELSE IF uc = visit THEN
            IF C > 0 THEN
                C ← C - 1
                Bike ← Bike + 1
                Ac ← Ac + 1
            ELSE
                Rj ← Rj + 1
            ENDIF
        ENDIF
    ENDIF

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ELSEIF $vt = \text{Car}$ THEN

IF $vc = \text{Fac}$ THEN

IF $A > 0$ THEN

$A \leftarrow A - 1$

$\text{Car} \leftarrow \text{Car} + 1$

$Ac \leftarrow Ac + 1$

ELSE

$Rj \leftarrow Rj + 1$

ENDIF

ELSEIF $vc = \text{Stud}$ THEN

IF $B > 0$ THEN

$B \leftarrow B - 1$

$\text{Car} \leftarrow \text{Car} + 1$

$\text{RAC} \leftarrow \text{AC} + 1$

ELSE

$Rj \leftarrow Rj + 1$

ELSEIF $vc = \text{visit}$ THEN

IF $C > 0$ THEN

$C \leftarrow C - 1$

$\text{Car} \leftarrow \text{Car} + 1$

$\text{AC} \leftarrow \text{AC} + 1$

ELSE

$Rj \leftarrow Rj - 1$

ENDIF

ENDIF

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ELSE IF v = van THEN
    IF VC = Fail THEN
        IF A > 1 THEN
            A ← A - 2
            v-an ← v-an + 2 |
            AC ← AC + 1
        ELSE
Rj ← Rj + 1
END IF
ELSE IF VC = Stud OR VC = visit THEN
    IF C > 0 THEN
        C ← C - 2
        v-an ← v-an + 1
        AC ← AC + 1
    ELSE
        Rj ← Rj + 1
    ENDIF
ENDIF
ENDIF
ENDIF
ENDIF
ENDIF
OUTPUT NG N-1
END WHILE
OUTPUT "Capacity Zone A:", A, "Zone B", B, "Zone C", C
OUTPUT "Dikes:", Dike, "Cars:", Car, "Vans:", Vans
OUTPUT "Accepted", AC, "Rejected" RJ
OUTPUT "Processed", N

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